

Chapter 2 Probability

What is Probability

Refers to the study of randomness and uncertainty.

In any situations where a number of possible outcomes may occur. The theory of probability provides methods for quantifying changes.

2.1 Sample Spaces and Events

Experiment

Any action or process whose outcome is subject to uncertainty. (toss a coin or several times)

Sample space

Denoted by S , is the set of all possible outcomes of the experiment.

For example, two possible outcomes of fuse D & N, two fuses in sequence

$$S = \{DD \ DN \ ND \ NN\}$$

Events

We will be interested in not only the individual outcomes of S but also various collections of outcomes from S .

An event is any collection(subset) of outcomes contained in the sample space S .

- Simple Event(elementary event)

An event consists of exactly one outcome.

- Compound Event

An event consists of more than one outcome.

For example, with the sample space $S = \{1, 2, \dots, 18\}$, "The outcome is a prime number" is compound event(the subset is $E = \{2, 3, 5, 7, 11, 13, 17\}$) whereas "The outcome is the square of 4" is simple event(the subset is $E = \{16\}$).

2.2 Axioms, Interpretations, and Properties of Probability

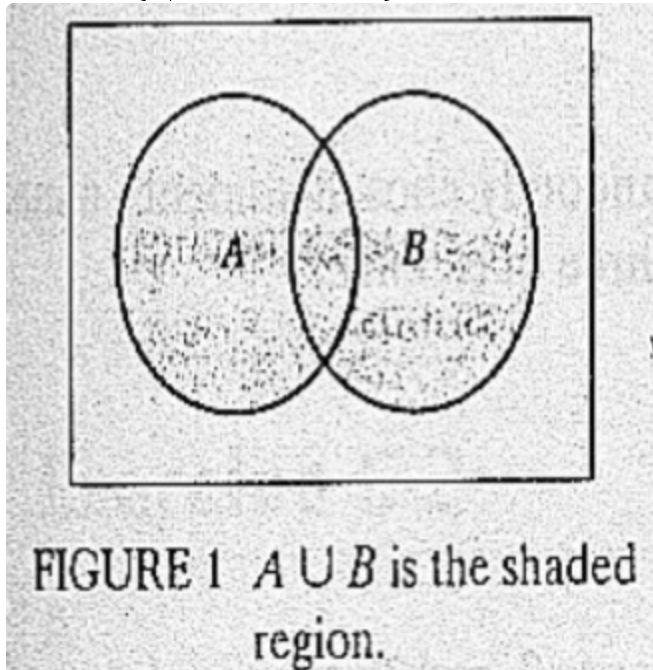
Some Relations from Set Theory

An event is just a set so relationship and results from elementary set theory can be used to study events.

- Union

The set of all outcomes in at least one of the events A and B , denoted by $A \cup B$, read "A or B".

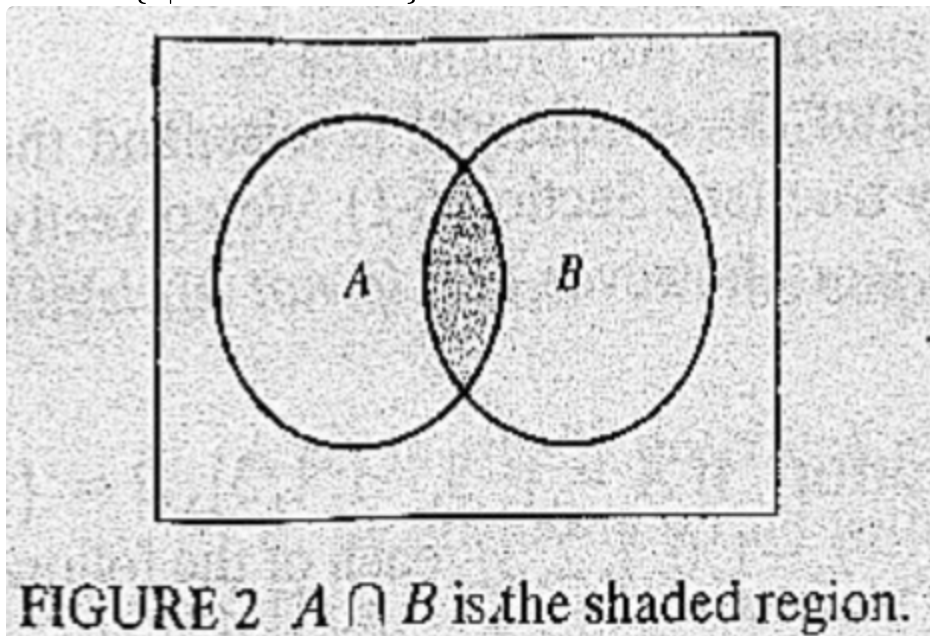
$$A \cup B = \{x | x \in A \text{ or } x \in B\}$$



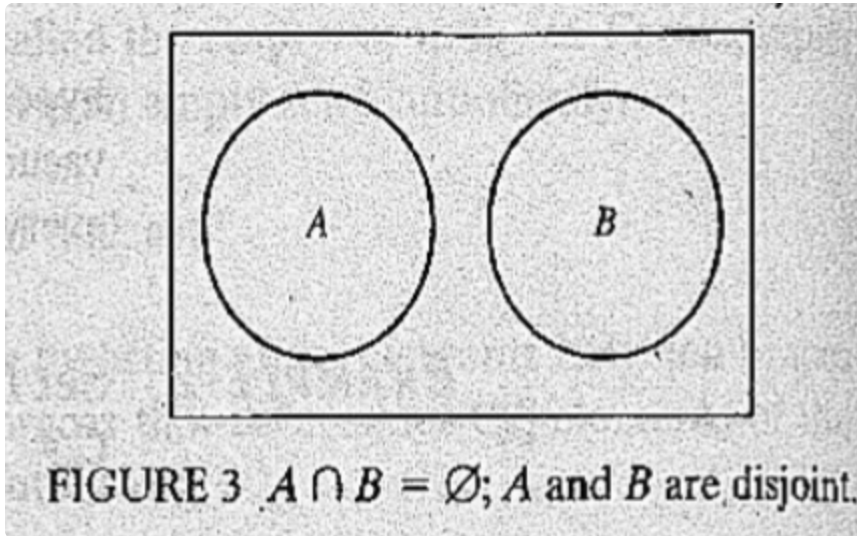
- Intersection

The set of all outcomes that are in both A and B , denoted by $A \cap B$, read "A and B"

$$A \cap B = \{x | x \in A \text{ and } x \in B\}$$



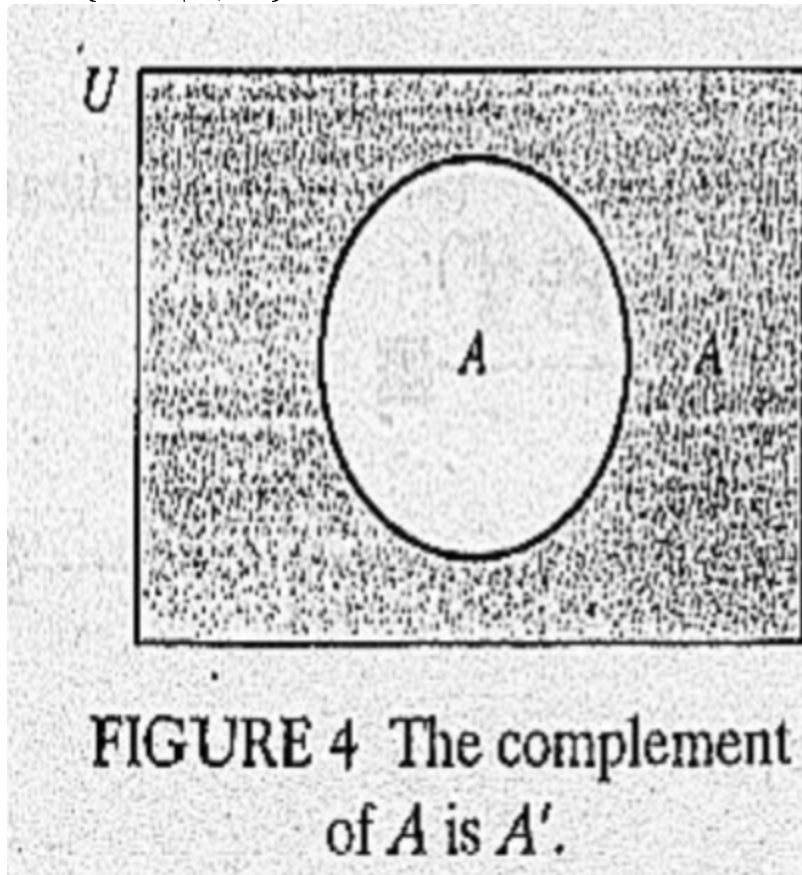
- Disjoint: $A \cap B = \emptyset$ (mutually exclusive or disjoint events)



- Complement

Denoted by A' . The set of all outcomes in S that are not contained in A .

$$A' = \{x \in U | x \notin A\}$$



Axioms, Interpretations, and Properties of Probabilities

Given an experiment and a sample space S , the object of probability is to assign each event A a number $P(A)$.

- Axiom 1: for any event A , $P(A) \geq 0$.

- Axiom 2: $P(S) = 1$.

- Axiom 3:

If $A_1, A_2, \dots$ is an **finite** collection of **mutually exclusive events**, then

$$P(A_1 \cup A_2 \cup \dots) = \sum_{i=1}^k P(A_i)$$

If it is an **infinite collection** of **mutually exclusive events**, then

$$P(A_1 \cup A_2 \cup \dots) = \sum_{i=1}^{\infty} P(A_i)$$

Proposition: $P(\emptyset) = 0$.

Two special events

- Impossible event

The event contains **no simple event**. ($P(A) = 0$)

- Certain event

The event contains **all simple events**. ($P(B) = 1$)

Interpreting Probability

Methods for assigning appropriate/correct probability

- Based on **repeatedly experiments(objective)**
- Based on some **reasonable assumption** or **prior information(subjective)**

Relative frequency vs. Probability

As n gets arbitrarily large, $n(A)/n$ approaches a limit value, the limit value is $P(A)$.

Property 1

For any event A , $P(A) = 1 - P(A')$.

Property 2

If A and B are **mutually exclusive**, then $P(A \cap B) = 0$.

Property 3

For **any two events** A and B ,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

For any n events $A_1, \dots, A_n$

$$\left| \bigcup_{i=1}^n A_i \right| = \sum_{k=1}^n (-1)^{k+1} \sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}|$$

Determining Probabilities Systematically

- First determine probability $P(E_i)$ for all simple events.
- The probability of any compound event A is computed by adding together the $P(E_i)$'s for all E_i 's in A .

Equally Likely Outcomes

Equal probabilities to all N simple events

With $p = P(E_i)$ for every i , then $p = \frac{1}{N}$.

Consider an event A , with $N(A)$ denoting the number of outcomes containing in A , then

$$P(A) = \sum_{E_i \text{ in } A} \frac{1}{N} = \left(\frac{N(A)}{N} \right)$$

2.3 Counting Techniques

Addition Principle

To do one thing and completed it can have n -class way, for each class there are m_i different ways. So total $N = \sum_{i=1}^n m_i$ different ways

Or, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$.

If A and B are disjoint, then $n(A \cup B) = n(A) + n(B)$.

Multiplication principle

To do one thing and completed it needs to divided into n steps, for each step there are m_i different ways. So total $N = m_1 m_2 \dots m_n$ different ways

- Ordered pair

If O_1 and O_2 are different objects, then the pair (O_1, O_2) is different from the pair (O_2, O_1)

Permutation

An arrangement of the objects in a specific order without repetition.

- Number of Permutations of n Objects

The number of permutations of n distinct objects without repetition denoted by $P_{n,n}$ is

$$P_{n,n} = n(n-1) \cdots 2 \cdot 1 = n!$$

- Permutation of n Objects Taken r at a time

A permutation of a set of n distinct objects taken r at a time without repetition is an arrangement of r of n objects in a specific order.

$$P_{n,r} = \frac{n!}{(n-r)!}$$

Combination

The arrangement of the element in the subset **does not matter**.

Combination of n objects taken r at a time

$$C_{n,r} = \frac{P_{n,r}}{r!} = \frac{n!}{r!(n-r)!}$$

2.4 Conditional Probability

We will use $P(A|B)$ to represent **conditional probability of A given B** . B is the **conditioning event**.

Definition of Conditional Probability

For any event A and B with $P(B) > 0$, the conditional probability of A given B is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Given that B has occurred, the relevant sample space for $P(A|B)$ is **no longer S** but consists of **outcomes in B** . A has occurred **if and only if** one of the outcomes in the **intersection** occurred.

- The Multiplication Rule

$$P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$$

Note:

- For three events,

$$P(A_1 \cap A_2 \cap A_3) = P(A_1|A_2 \cap A_3)P(A_2 \cap A_3) = P(A_1|A_2 \cap A_3)P(A_2|A_3)P(A_3)$$

- For n events,

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_n|A_1 \cap \cdots \cap A_{n-1})P(A_{n-1}|A_1 \cap \cdots \cap A_{n-2}) \cdots P(A_2|A_1)P(A_1)$$

The Law of Total Probability

$$P(B) = P(B \cap A) + P(B \cap A') = P(B|A)P(A) + P(B|A')P(A')$$

- Partition of S

Definition: We say the events $A_1, \dots, A_n$ represent a **partition** of the sample space S if

- $A_i \cap A_j = \emptyset$ for all $i \neq j$
- $\bigcup_{i=1}^n A_i = S$
- $P(A_i) > 0$ for all $i = 1, 2, \dots, n$.

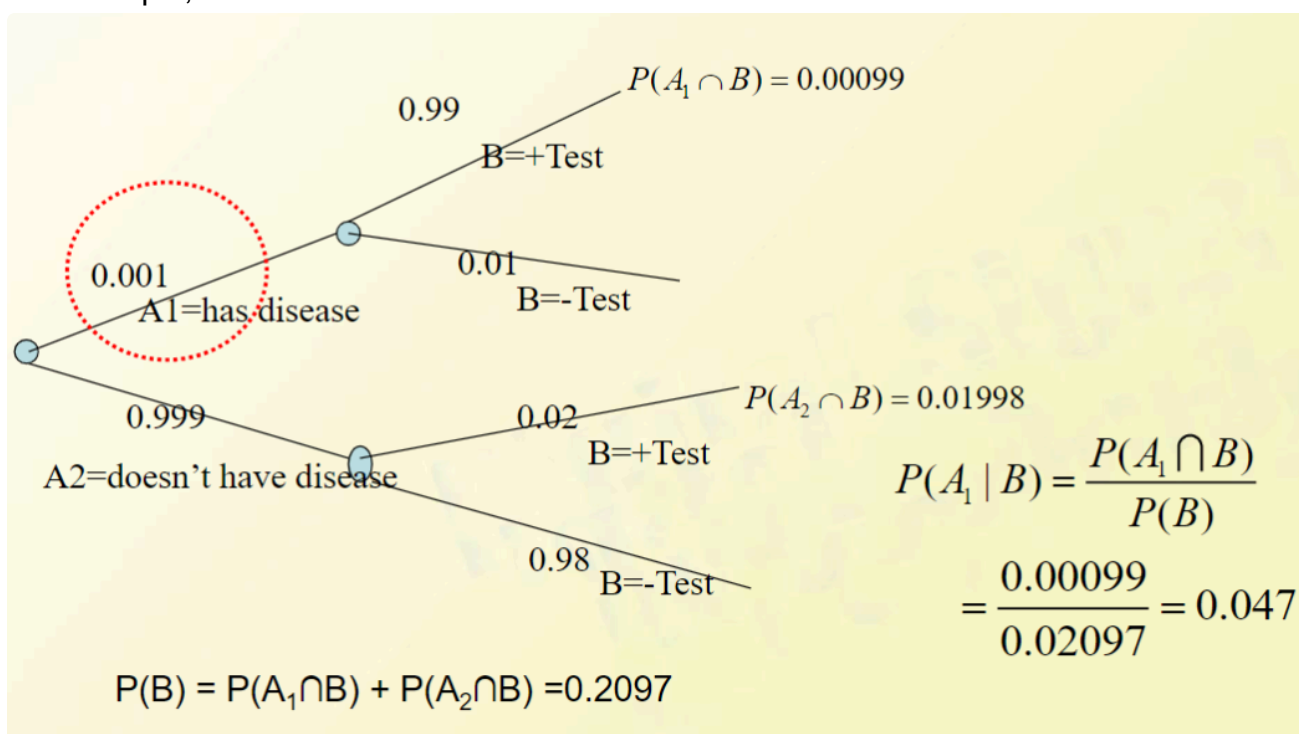
- The Law of Total Probability(**general cases**)

Let $A_1, \dots, A_k$ be partition of S , then for any other event B ,

$$P(B) = \sum_{i=1}^k P(A_i \cap B) = \sum_{i=1}^k P(A_i)P(B|A_i)$$

- Probability Tree

For example,



Bayes' Theorem

Let $A_1, \dots, A_k$ be partition of S with $P(A_i) > 0$, then for any other event B for which $P(B) > 0$,

$$P(A_j|B) = \frac{P(A_j \cap B)}{P(B)} = \frac{P(A_j)P(B|A_j)}{P(B)} = \frac{P(A_j)P(B|A_j)}{\sum_{i=1}^k P(A_i)P(B|A_i)}$$

where $j = 1, 2, \dots, k$.

Useful for Machine Learning, Artificial Intelligence.

2.5 Independence

Definition

Two events A and B are independent if $P(A|B) = P(A)$ and are dependent otherwise.

Note:

- If $P(A|B) = P(A)$, we have $P(A \cap B) = P(A)P(B)$ and $P(B|A) = P(B)$.
- If A and B are independence, so are the following pairs of events:
 - A', B
 - A, B'
 - A', B'
- If two events are mutually exclusive, they cannot be independent.

Proposition 1

A and B are independent if and only if

$$P(A \cap B) = P(A)P(B)$$

Independence of more than two events

- Mutually Independent

Events $A_1, \dots, A_n$ are mutually independent if for every $k = 2, 3, \dots, n$ and every subset of indices $i_1, i_2, \dots, i_k$,

$$P(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_k}) = P(A_{i_1})P(A_{i_2}) \dots P(A_{i_k})$$

That is, for any number of events, they satisfy the equation.